Conditional Diffusion Models for Camouflaged and Salient Object Detection

Ke Sun, Zhongxi Chen, Xianming Lin, Xiaoshuai Sun, *Member, IEEE*, Hong Liu, and Rongrong Ji, *Senior Member, IEEE*

***Abstract*—Camouflaged Object Detection (COD) poses a significant challenge in computer vision, playing a critical role in applications. Existing COD methods often exhibit challenges in accurately predicting nuanced boundaries with high-confidence predictions. In this work, we introduce CamoDiffusion, a new learning method that employs a conditional diffusion model to generate masks that progressively refine the boundaries of camouflaged objects. In particular, we first design an adaptive transformer conditional network, specifically designed for integration into a Denoising Network, which facilitates iterative refinement of the saliency masks. Second, based on the classical diffusion model training, we investigate a variance noise schedule and a structure corruption strategy, which aim to enhance the accuracy of our denoising model by effectively handling uncertain input. Third, we introduce a Consensus Time Ensemble technique, which integrates intermediate predictions using a sampling mechanism, thus reducing overconfidence and incorrect predictions. Finally, we conduct extensive experiments on three benchmark datasets that show that: 1) the efficacy and universality of our method is demonstrated in both camouflaged and salient object detection tasks. 2) compared to existing state-of-the-art methods, CamoDiffusion demonstrates superior performance 3) CamoDiffusion offers flexible enhancements, such as an accelerated version based on the VQ-VAE model and a skip approach.**

***Index Terms*—Camouflaged object detection (COD), diffusion model, salient object detection, transformer network.**

自适应交换器条件网络，集成去噪网络，模型迭代细化
方差噪声调度和结构破坏，处理不确定输入，提高准确性
共识时间集成技术，整合中间预测，减少过度以及不正确预测

Received 9 April 2024; revised 29 October 2024; accepted 31 December 2024. Date of publication 8 January 2025; date of current version 6 March 2025. This work was supported in part by the National Science Fund for Distinguished Young Scholars under Grant 62025603, in part by the National Natural Science Foundation of China under Grant U21B2037, Grant U22B2051, Grant U23A20383, Grant 62176222, Grant 62176223, Grant 62176226, Grant 62072386, Grant 62072387, Grant 62072389, Grant 62002305, and Grant 62272401, and in part by the Natural Science Foundation of Fujian Province of China under Grant 2021J06003 and Grant 2022J06001. Recommended for acceptance by M. Long. *(Corresponding author: Hong Liu.)*

Ke Sun, Zhongxi Chen, and Xianming Lin are with the Media Analytics and Computing Laboratory, Department of Artificial Intelligence, School of Informatics, Xiamen University, Xiamen 361005, China.

Xiaoshuai Sun and Rongrong Ji are with the Media Analytics and Computing Laboratory, Department of Artificial Intelligence, School of Informatics, Xiamen University, Xiamen 361005, China, and also with the Institute of Artificial Intelligence, Xiamen University, Xiamen 361005, China (e-mail: xssun@xmu.edu.cn; rrji@xmu.edu.cn).

Hong Liu is with the Institute for Datability Science, Osaka University, Osaka 565-0871, Japan (e-mail: hliu@ids.osaka-u.ac.jp).

Our codes and models are available at https://github.com/Rapisurazurite/CamoDiffusion.

This article has supplementary downloadable material available at https://doi.org/10.1109/TPAMI.2025.3527469, provided by the authors.

Digital Object Identifier 10.1109/TPAMI.2025.3527469

I. Introduction

Camouflage serves as a critical survival mechanism in the natural world, enabling organisms to blend into their environment to evade predators and stealthily approach prey. The study of camouflaged object detection (COD) has thus become a crucial research area in artificial intelligence [1], [2], with applications ranging from biodiversity conservation [3], medical image diagnostics [4], to manufacturing defect detection [5].

The past few years have witnessed significant advances in the field of COD, with numerous representative approaches being introduced in the research area. However, as shown in Fig. 1, recent representative approaches, such as SINet-V2 [1] and ZoomNet [6], face two challenges when dealing with complex camouflaged objects. The first is the ambiguous boundaries between the target objects and their background [2]. Due to the nature of camouflage, in which objects seamlessly blend with their surroundings, accurately delineating the boundaries becomes a highly intricate task. The subtle visual cues and intricate patterns exhibited by camouflaged objects make it difficult for existing methods to precisely distinguish the object from its background, resulting in blurred and imprecise boundaries. The second limitation is the issue of over-confident predictions [7]. Due to the center-biased datasets commonly used in COD [8], models often misclassify non-camouflaged regions as camouflaged objects, resulting in overconfident predictions that lead to false positives and diminish the overall accuracy. These challenges are not unique to COD but also exist in Salient Object Detection (SOD), where the objective is to detect visually distinct objects in an image. Both tasks share a need for effective boundary detection and feature differentiation, as camouflaged objects in COD and salient objects in SOD often require nuanced contextual understanding to be properly segmented [9], [10].

物体与周围的模糊边界，边界划分困难。

过于自信，预测扩大伪装范围

Given such specific challenges posed by COD and SOD, we adopt the diffusion model paradigm [11] as a fitting solution. Diffusion models possess impressive generative capabilities with a good awareness of conditions. Their iterative noise reduction mechanism obviates the need for intricate refinement modules within prevailing models. This mechanism enables a progressive differentiation between boundary nuances and contextual surroundings, fostering a more holistic comprehension. The inherent stochastic sampling process enables the generation of multiple predictions and the evaluation of segmentation uncertainty, which mitigates the risk of erroneous confidence in the model's results.

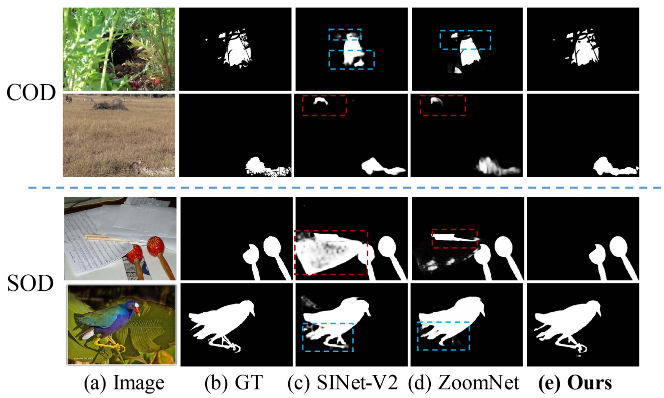


Fig. 1. The visual results of various methods on COD and SOD tasks. Blue boxes indicate the boundary-context distinction of objects, and red boxes point to overconfident mis-segmentations, where methods incorrectly segment parts of the image with high certainty. Our approach effectively addresses these issues, reducing instances of mis-segmentation and improving the clarity of boundary-context distinctions.

扩散模型处理：新型去噪网络细化掩模，新型自适应交换器条件网络捕捉边界细节，信噪比方差噪声调度的高效，结构腐败策略用于去噪提高预测偏差能力。

This paper presents a novel approach, called **CamoDiffusion**, which tackles the COD task through the lens of conditional diffusion models. CamoDiffusion generates the mask by progressive denoising a noise map under a conditional context obtained from the camouflaged image, which achieves an effective and accurate COD method with four main innovations detailed below:

First, based on the basic principles of DDPM, we present a new UNet-based denoising network (DN) that aims to refine the saliency mask through an iterative conditional denoising process. To obtain good visual context conditions, we design a novel **adaptive transformer conditional network** (ATCN) that builds on a pyramid visual transformer along with a newly designed directional guidance mechanism, with the aim of effectively capturing the subtle details and boundaries of camouflaged objects.

Direct adoption of the standard diffusion model training process leads to performance degradation. To overcome this, we present two techniques during training as follows: 1) We show that a variance noise schedule in terms of signal-to-noise ratio works well in the COD task: The noise schedule should be adjusted to generate a high-resolution camouflage mask. 2) We introduce a structure corruption strategy that exposes the model to corrupted object contours, aiding the model in learning to denoise at the structural level and improving its ability to correct biases in early predictions.

Third, since CamoDiffusion generates the saliency mask by iteratively denoising a noise map, we observe that the generated masks exhibit incorrect intermediate predicted results due to the overconfidence issue. To overcome this and encourage generating correct final generated masks, we proposed a Consensus Time Ensemble (CTE) sampling technique. CTE fully utilizes the valuable predicted information generated during the denoising process and integrates them together through a newly designed voting mechanism.

Finally, to ensure the practicality and efficiency of our approach, we introduce some acceleration strategies for CamoDiffusion. First, we use the ATCN-Skip strategy, which skips redundant computations during the diffusion steps. Second, we apply the diffusion process in the latent space of a pretrained Auto-Encoder [12], allowing our method to maintain precision and flexibility even on limited computational resources.

The experiments show that our method achieves a marked improvement over previous state-of-the-art approaches, highlighting its versatility and robustness.

Our main contributions are summarized as follows:

1) We treat both COD and SOD tasks via a conditional diffusion framework for mask prediction.
2) We introduce CamoDiffusion with a specially designed network structure and learning strategies.
3) Our method achieves state-of-the-art performance on three COD datasets and three SOD datasets.

This work extends from our previous work in [13]. The six new contributions include: (1) We propose a new accelerated version of our method, along with a discussion and experimental analysis; (2) We conduct additional quantitative and qualitative experiments on the SOD task to further validate our approach; (3) We offer a detailed description of the training and sampling processes employed in our method; (4) We include a description and visualization of the Structure Corruption and CTE modules utilized in our framework; (5) We perform an extensive set of ablation studies to investigate the effectiveness of our proposed method; (6) We provide an in-depth analysis of the backbone, sampling technique, structure corruption, and noise schedule components.

II. Method

This section presents our CamoDiffusion framework, which progressively refines predictions using image priors at each step, as shown in Fig. 2. Built on the DDPM framework, CamoDiffusion iteratively denoises a noise map to generate the COD mask, conditioned on features extracted by a proposed conditional network.

The training process begins by initializing the Adaptive Transformer Conditional Network (ATCN) and the Denoising Network (DN). During training, batches of input images and ground truth masks undergo structure corruption to enhance boundary complexity, followed by noise addition using an SNR-based variance schedule. The ATCN extracts multi-scale features from both images and noisy masks, utilizing Zero Overlapping Embedding (ZOE) and Time Token Concatenation (TTC) modules. These features are then passed to the DN, which progressively refines the masks, enhancing boundary precision at each step.

We start by explaining the background and notation in Section II-A. Then, we dive into the newly designed neural network in Section II-B. Finally, Section II-C and II-D detail our unique training and sampling strategies, respectively.

A. Background and Notation

Our CamoDiffusion is based on the typical conditional diffusion model [11], which includes a forward process, where a

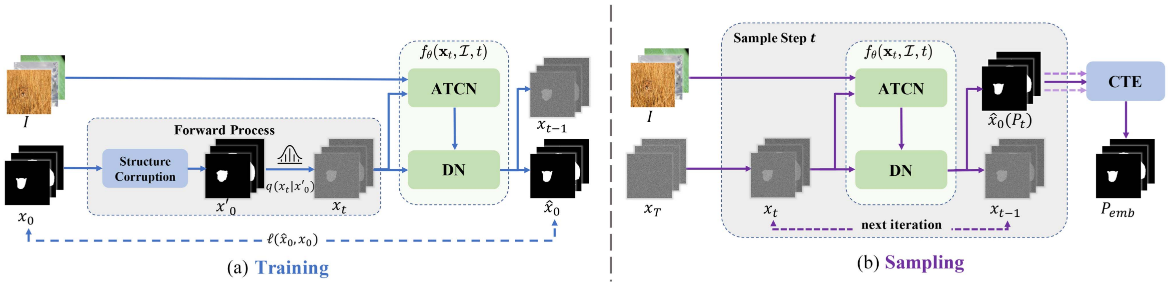


Fig. 2. The overall framework of CamoDiffusion, which includes an Adaptive Transformer Conditional Network (ATCN), and a Denoising Network (DN). In training, the ground truth (gt) $\mathbf{x}_0$ undergoes Structure Corruption and Gaussian noise to generate the noised mask $\mathbf{x}_t$ at time t. Subsequently, our model predicts the denoised mask $\hat{\mathbf{x}}_0$ and is trained to minimize the loss between the prediction and the gt. During sampling, our model denoises the random noise $\mathbf{x}_T$ for the T steps. All predictions $\{P_t\}_{t=1}^{T}$ generated in the reverse process are aggregated by the Consensus Time Ensemble (CTE), resulting in a more reliable outcome P_{emb}.

mask is gradually noised, and a reverse process, where the noise is transformed back to the target distribution.

Given a training sample $\mathbf{x}_0 \sim q(\mathbf{x}_0)$, where $\mathbf{x}_0$ represents the ground truth mask for camouflaged object, the noised versions $\{\mathbf{x}_t\}_{t=1}^{T}$ are obtained according to the following Markov process:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I}), \tag{1}$$

where t runs from 1 to T, and variance is controlled by noise schedule $\beta_t \in (0, 1)$. And the marginal distribution of $\mathbf{x}_t$ can be described as:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \tag{2}$$

where $\bar{\alpha}_t = \prod_{i=1}^{T} \alpha_t, \alpha_t = 1 - \beta_t$.

Starting from $p(\mathbf{x}_T) = \mathcal{N}(\mathbf{x}_T; \mathbf{0}, \mathbf{I})$, the reverse process uses a neural network f_θ to create a sequence of incremental denoising operations to obtain back the clean mask. This network learns the reverse distribution as follows:

$$p(\mathbf{x}_{t-1}|\mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)). \tag{3}$$

Although direct modeling $\mathbf{x}_{t-1}$ is possible, following the approach of [11], we find that maintaining a consistent output space for the network improves performance.

To this end, we opt to learn a network $f_\theta(\mathbf{x}_t, \mathcal{I}, t)$ to predict the denoised mask $\hat{\mathbf{x}}_0$, conditional on the image $\mathcal{I}$. In practice, $\Sigma_\theta(\mathbf{x}_t, t)$ is set to $\sigma_t^2 = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t$ and $\mu_\theta(\mathbf{x}_t, t)$ can be expressed as follows:

$$\mu_\theta(\mathbf{x}_t, t) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\hat{\mathbf{x}}_0. \tag{4}$$

Here, $\hat{\mathbf{x}}_0$ is predicted by $f_\theta(\mathbf{x}_t, I, t)$, which consists of a new ATCN and DN as described in the following section. Then we learn the model f_θ with the following loss function:

$$\ell(\hat{\mathbf{x}}_0, \mathbf{x}_0) = \ell_{IoU}^{w}(\hat{\mathbf{x}}_0, \mathbf{x}_0) + \ell_{BCE}^{w}(\hat{\mathbf{x}}_0, \mathbf{x}_0), \tag{5}$$

where ℓ_{IoU}^{w} and ℓ_{BCE}^{w} represent weighted intersection-over-union (IoU) loss and weighted binary cross entropy (BCE) loss, respectively.

B. Architecture Design

Our model diverges from the traditional segmentation approach by leveraging conditional diffusion models for prediction generation. Fig. 3 shows the pipeline of our designed network. We utilize ATCN to capture hierarchical image features as conditions, which are subsequently incorporated into DN. The design specifics of these networks are elaborated in subsequent sections.

1) Adaptive Transformer Conditional Network (ATCN): The primary objective of our ATCN is to extract rich visual information from the COD task and identify the most discriminative features between the camouflaged object and the background. To achieve this, we employ the Pyramid Vision Transformer V2 (PVT) [14] as our base architecture, which enables multi-scale feature extraction. However, the camouflaged nature of objects significantly complicates the extraction of discriminative image features, which can adversely affect the results.

To address this challenge, we propose the *Zero Overlapping Embedding* (ZOE) module, which integrates a roughly predicted mask from an earlier step as a directional cue, focusing the network's attention on particular areas. This strategy uncovers the detailed shapes and edges of camouflaged objects, enhancing the network's ability to extract discriminative features.

Furthermore, to improve the adaptability of the features throughout the denoising phases, we introduce the *Time Token Concatenation* (TTC) module, which incorporates the variable in the denoising step t into the conditional network. This enables the network to dynamically adapt its feature extraction based on the current denoising step, leading to more effective and tailored feature representations.

The framework of the ATCN is shown in Fig. 3(a), and the detailed descriptions of the ZOE and TTC modules are provided below:

ZOE Module: To integrate the noise mask $\mathbf{x}_t$ into the PVT while preserving the original transformer structure and pre-training parameters, we introduce Zero Overlapping Embedding (ZOE) at the first layer of ATCN, replacing the traditional Overlapping Embedding (OE). ZOE employs an additional convolution layer, initialized with zeros, to subtly blend $\mathbf{x}_t$ into the network.

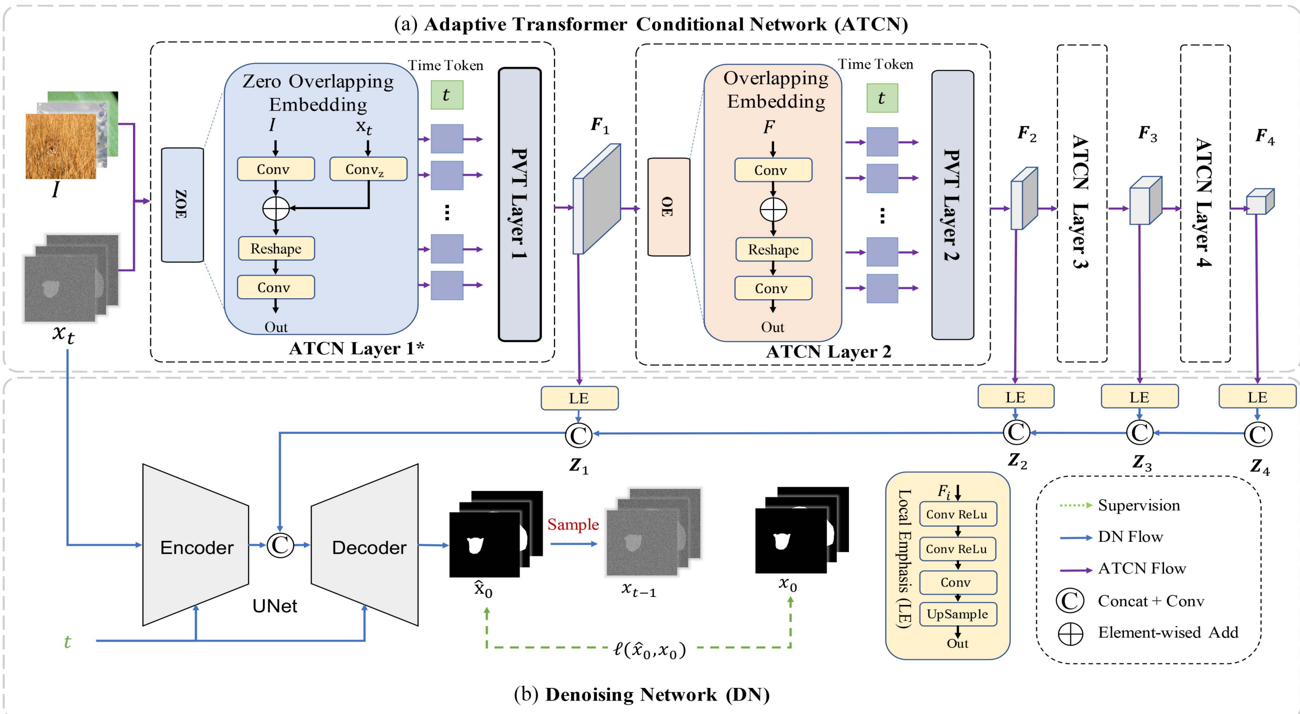


Fig. 3. The architecture design of our CamoDiffusion, which includes an ATCN for multi-scale feature extraction and a DN for generating precise mask predictions from noised inputs. The notation "*" indicates modifications to the PVT layer by our Zero Overlapping Embedding (ZOE) technique. The symbol 'C' represents a concatenating operation, followed by a 1×1 convolution to align the channel dimensions.

Algorithm 1: Training.

Require: Image and mask distribution $q(\mathcal{I})$, $q(\mathbf{x}_0|\mathcal{I})$
repeat
  $t \sim \text{Uniform}(\{1, \ldots, T\})$
  $\mathcal{I} \sim q(\mathcal{I})$
  $\mathbf{x}_0 \sim q(\mathbf{x}_0|\mathcal{I})$
  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  $\mathbf{x}'_0 = \text{structureCorrupt}(\mathbf{x}_0)$
  $\mathbf{x}_t = \alpha_t \mathbf{x}'_0 + \sigma_t \epsilon$
  Take gradient descent step on $\nabla_\theta \ell(\mathbf{x}_0, f(\mathbf{x}_t, \mathcal{I}, t))$
until converged

Algorithm 2: Sampling.

Require: Image $\mathcal{I}$
$\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
$\mathbf{x}_p = []$
for $t = T, \ldots, 1$ **do**
  $\hat{\mathbf{x}}_0 = f_\theta(\mathbf{x}_t, \mathcal{I}, t)$
  $\text{appendToHistory}(\mathbf{x}_p, \hat{\mathbf{x}}_0)$ //Append $\hat{\mathbf{x}}_0$ to $\mathbf{x}_p$
  $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $\mathbf{z} = \mathbf{0}$
  $\mathbf{x}_{t-1} = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\hat{\mathbf{x}}_0 + \sigma_t \mathbf{z}$
end for
$\mathbf{p} = \text{consensusTimeEnsemble}(x_p)$
return p

This method ensures that the position encoding is unaffected during initialization, allowing the merging of noise masks to be controlled without modifying the base transformer architecture.

The mathematical representation of the ZOE of the i-th layer is given by:

$$\mathbf{emb_i} = \begin{cases} \text{LN}(\mathtt{R}(\text{Conv}(\mathcal{I}) + \text{Conv}_z(\mathbf{x}_t))), & i = 1, \\ \text{LN}(\mathtt{R}(\text{Conv}(\mathbf{F}_{i-1}))), & i \neq 1, \end{cases} \tag{6}$$

where Conv$(\cdot)$ signifies the original convolution operation, Conv$_z(\cdot)$ represents a convolution operation whose weight and bias are both initialized to zeros, $\mathtt{R}(\cdot)$ is a function that transforms the feature map produced by the convolution into tokens, and LN$(\cdot)$ refers to layer normalization [15].

TTC Module: In addition to handling the noisy mask, our ATCN is designed to automatically tailor the condition's characteristics based on the temporal stage. So we introduce the TTC module as follows:

$$\mathbf{F}_i = \mathtt{R}^{-1}(\text{FFN}(\text{MHA}([\mathbf{t}; \mathbf{emb_i}]))), \tag{7}$$

where $\mathbf{t}$ represent the time token, $[\cdot;\cdot]$ indicates the concatenation operation, $\mathtt{R}^{-1}$ is the inverse process that converts tokens back into a feature map, the MHA stands for multi-head self-attention, and the FFN is the feed-forward neural network. These advancements set our approach apart from traditional diffusion models, which in previous studies [16], [17] depended only on image input, reusing identical image features during reverse processes.

2) Denoising Network: The Denoising Network, as depicted in Fig. 3(b), is designed to generate denoised mask predictions $\hat{\mathbf{x}}_0$ and $\mathbf{x}_{t-1}$, adhering to the diffusion model. The Encoder $\mathrm{Enc}(\cdot)$ and Decoder $\mathrm{Dec}(\cdot)$ work together to extract detailed information from $\mathbf{x}_t$ and make predictions conditional on the feature map $\mathbf{Z}_1$, utilizing the following equations:

$$\hat{\mathbf{x}}_0 = \mathrm{Dec}(\mathrm{Conv}([\mathbf{Z}_1, \mathrm{Enc}(\mathbf{x}_t)])), \tag{8}$$

$$\mathbf{x}_{t-1} = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\hat{\mathbf{x}}_0 + \sigma_t \mathbf{z} \tag{9}$$

where $\mathrm{Enc}(\cdot)$ and $\mathrm{Dec}(\cdot)$ mainly consist of several convolutional blocks. Here, $\mathrm{Enc}(\cdot)$ facilitates downsampling through stride convolutions, whereas the $\mathrm{Dec}(\cdot)$ employs PixelShuffle [18] for upsampling.

The Denoising Network integrates the time variable t using Adaptive Group Normalization (AdaGN) [19]. Specifically, $\mathrm{AdaGN}(h, y) = y_s \mathrm{GroupNorm}(h) + y_b$, where h represents intermediate activations within the convolutional layers, and y_s, y_b are derived from a linear projection of the time embedding. Rather than relying on complex, hierarchically refined decoders, our approach yields satisfactory outcomes through a straightforward structure, benefiting from the iterative denoising process of diffusion models.

Specifically, the feature sets $\{\mathbf{F}_i\}_{i=1}^4$ are upscaled to a uniform size via the Local Emphasis (LE) module [20]:

$$\mathrm{LE}(\mathbf{F}_i) = \mathrm{Up}(\mathrm{CR}(\mathrm{CR}(\mathbf{F}_i))), \tag{10}$$

where $\mathrm{Up}(\cdot)$ is the bilinear interpolation and $\mathrm{CR}(\cdot)$ represents the convolution operation followed by ReLU activation. The features undergo gradual aggregation as follows:

$$\mathbf{Z}_i = \mathrm{Conv}([\mathbf{Z_{i+1}}, \mathrm{LE}(\mathbf{F}_i)]), \quad i \in \{3, 2, 1\}, \tag{11}$$

where $\mathbf{Z_4} = \mathrm{LE}(\mathbf{F}_4)$. Ultimately, a compact encoder and decoder are employed to denoise $\mathbf{x}_t$, guided by the condition $\mathbf{Z}_1$ and the time step t.

C. Training Strategy

During training, we start the diffusion process from the ground truth (GT) mask towards the noisy mask, which serves to train our conditional denoising network to approximately invert this process. Despite the effectiveness of training, we face specific challenges: restoring a clear mask from masks with a low Signal-to-Noise Ratio (SNR). Furthermore, the model faces an additional challenge in its limited ability to accurately refine the contours of camouflaged objects. To overcome this, we introduce a *Structure Corruption* strategy, which aids the model in learning to denoise at the structural level. The whole training algorithm is detailed in Algorithm 1.

SNR-based Variance Schedule: On the one hand, previous work [17] shows that the masks have an excessive SNR in training phase, which poses a challenge for the model to recover the mask from low SNR inputs. On the other hand, COD poses a challenge due to its complex contexts and higher resolutions, which tend to present simpler cases during training. So we use an SNR-based variance noise schedule [21] in our training approach.

We use the variance schedule on the logarithmic scale by introducing an offset: $\mathrm{SNR}_{\text{shift}}(t) = \exp(\log \mathrm{SNR}(t) + \text{shift})$, where $\mathrm{SNR}(t) = \bar{\alpha}_t/(1-\bar{\alpha}_t)$ [22]. This strategy involves deliberately reducing the SNR of the input masks. By making this calculated adjustment, we aim to increase the difficulty of the training process. This encourages the model to conduct a more thorough exploration of the features and to adapt more effectively to the distinct challenges presented by COD task.

Structure Corruption: Traditional diffusion models apply a pixel-level corruption scheme [11], [23], [24] to directly create a noised mask from the GT mask, which can lead the model to mistakenly rely on the restored contours of the noised mask, thus limiting its ability to make corrections. The challenge of distinguishing boundaries of camouflaged objects often leads to significant inaccuracies in the model's initial segmentation predictions. Therefore, we introduce a new Structure Corruption in the diffusion process. This technique involves deliberately altering the contour of the GT by randomly removing parts of the contour and then applying Gaussian noise.

结构腐蚀：识别轮廓顶点，随机删除，移动替换剩余顶点，应用随机膨胀以及侵蚀，直到IOU降低到预期。

The structure corruption process is detailed as follows: It begins by identifying the contour vertices within the mask. Subsequent steps involve randomly deleting some vertices and then displacing the remaining ones through random movements. To further modify the mask, random dilation and erosion are applied until the Intersection over Union (IoU) decreases to a predetermined target level.

Specifically, we first identify contour vertices and apply regional sampling to remove consecutive vertices at a rate of 0.1. This is followed by random sampling of the remaining vertices at a 0.1 rate. The surviving vertices are then displaced based on their distance from the mask's center, with a small random movement rate. Finally, we apply random dilation and erosion operations to further alter the mask structure until a target IoU (randomly chosen between 0.8 and 1.0) is reached. This multi-step process ensures a comprehensive modification of the mask's structure while maintaining its overall shape, challenging the model to learn robust boundary detection.

This method enhances the model's capability to adjust for biases in earlier predictions, a vital step given the often unclear boundary contours of camouflaged objects. In our experiments, the target IoU is randomly chosen between 0.8 and 1.0 for each mask. Examples of masks after undergoing Structure Corruption are shown in Fig. 4.

D. Inference With CTE Sampling

During inference, the denoising model takes a sample $\mathbf{x}_T$ drawn from a standard normal distribution and applies incremental denoising over T steps to generate the final segmentation mask. This iterative denoising process steadily mitigates the divergence between the predicted mask and the ground truth, culminating in a more precise outcome.

In our CamoDiffusion, we also focus on mitigating the problem of overconfident predictions that lead to incorrect predicted masks (as Fig. 1 shown). Therefore, we introduce a **Consensus Time Ensemble** (CTE) approach based on the concept that

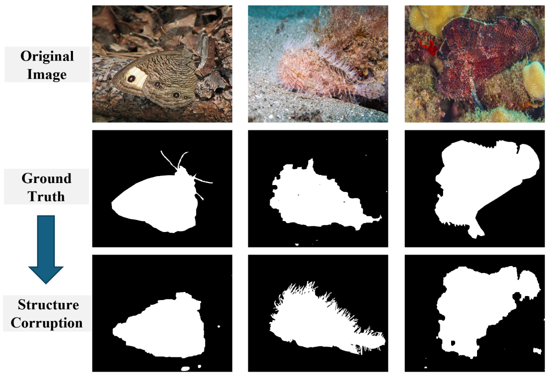


Fig. 4. Examples of masks after Structure Corruption.

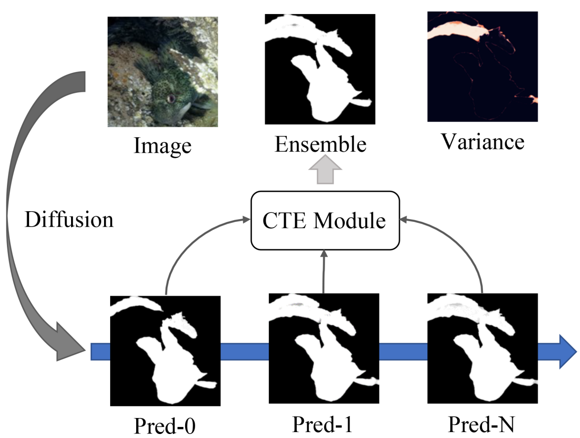


Fig. 5. Examples of our CTE strategy. The Variance map is obtained from pixel-wise variance across the different predictions and subsequently normalized to a range of 0-1.

the predictions generated during the denoising process carry valuable information.

CTE efficiently combines predictions from various sampling steps without additional computational demands. For each sampling stage at time t, the denoised image $\hat{\mathbf{x}}_0$ is represented as P_t. From multiple predictions $\{P_t\}_{t=1}^T$, binary masks $\{P_t^b\}_{t=1}^T$ are first generated using adaptive thresholds. These binary predictions $\{P_t^b\}_{t=1}^T$ collectively contribute to the formation of a candidate mask by voting on the position of each pixel. The probability value assigned to each selected pixel is calculated as the mean of all predictions as follows:

$$P_{emb} = \left\lfloor \frac{\sum_{t=1}^T P_t^b}{T} + \frac{1}{2} \right\rfloor * \text{mean}(P_t). \tag{12}$$

Moreover, our CTE is capable of producing multiple predictions by sampling from the mask distribution. This capability enables us to enhance the accuracy of the mask using ensemble methods or to gauge uncertainty by computing the variance, as illustrated in Fig. 5.

Practically, we sample the mask three times and utilize the CTE strategy to amalgamate these $3T$ predictions, thereby leveraging the strength of multiple inferences to refine the final mask outcome. The sampling strategy is detailed in Algorithm 2. The pseudocode for the entire process is provided in the appendix, available online.

E. Acceleration Strategy

ATCN-Skip: Our CamoDiffusion method involves using the ATCN as a conditional component to extract a rough outline of the camouflaged region from an input image. This initial outline is then refined through a multi-step diffusion process, which helps address issues such as imprecise boundaries and overconfident predictions. Previously, in each diffusion step, both the image I and the latent representation x_t were input into ATCN to compute the condition for refinement.

We argue that performing the same ATCN inference at every step is redundant, as the coarse localization can be adequately achieved during the initial step, and the subsequent refinement is primarily handled by the Diffusion Network (DN). Thus, we propose a skipping strategy named ATCN-Skip. As shown in Fig. 6, instead of re-computing Z_1 at every diffusion step, we reuse the Z_1 output from the previous step. Specifically, we skip ATCN computation in alternating steps, i.e., we use the ATCN output from the previous step for the current step, effectively halving the computation required by ATCN. This not only reduces the overall sampling time but also maintains competitive performance, as the feature maps do not change significantly between consecutive steps. The complete process is illustrated in the accompanying diagram, where the ATCN computation is skipped for every alternate step, and the $Z1$ output from step $t-1$ is reused for step t.

This ATCN-Skip strategy significantly reduces the computational load during inference, as it avoids redundant computation in the ATCN while retaining the effectiveness of the diffusion-based refinement. By adopting this skipping approach, we provide users with a flexible balance between accuracy and efficiency, making the CamoDiffusion model more adaptable to different scenarios and requirements.

Integrate with VAE: Inspired by [12], [25], we refine our model to function within the latent space for improved efficiency. Given the binary nature of high-resolution segmentation masks, we utilize a Vector Quantized Variational AutoEncoder (VQ-VAE) for transforming these mask images into a series of discrete latent variables.

First, the high-resolution binary mask $\mathbf{x}_t$ is passed through an encoder function E to obtain a lower-dimensional latent representation $\mathbf{z}_t$. The encoder function E is designed to capture the essential features and patterns present in the binary mask while reducing its spatial dimensions. Second, the diffusion process in the latent space follows the same principles as described in Section II-A, but operates on lower-dimensional latent representations instead of the original high-resolution masks. Third, performing the diffusion process in the latent space can significantly reduce the computational burden and memory requirements associated with processing high-resolution masks directly.

After the diffusion process is completed, the final denoised latent representation $\mathbf{z}_{t-1}$ is obtained. This denoised

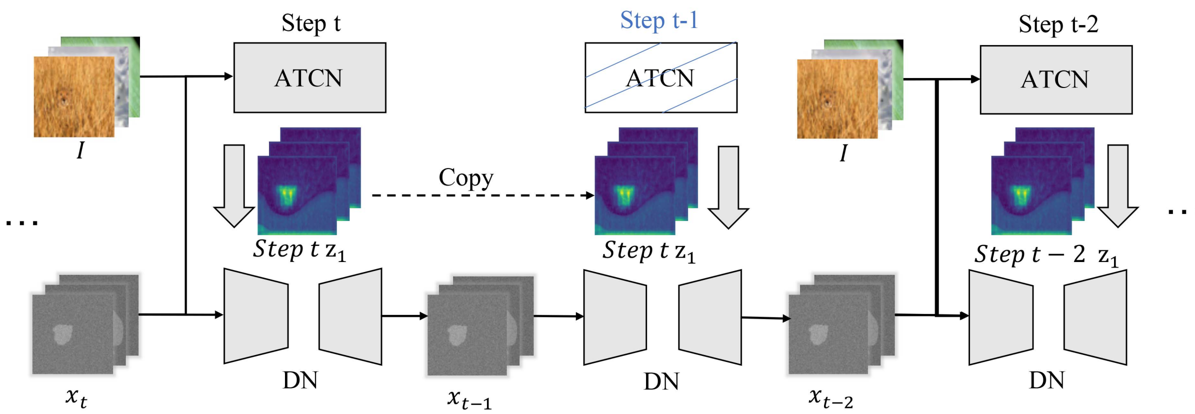


Fig. 6. Pipeline of our ATCN-Skip strategy, where ATCN computation is skipped every alternate time step to reduce computational redundancy.

latent representation captures the essential features and details of the mask in a compact form. To reconstruct the high-resolution binary mask from the denoised latent representation, we employ a decoder function D to map the latent representation back to the original high-resolution mask.

This approach allows us to leverage the computational efficiency and reduced complexity of operating in the latent space, as well as to apply sophisticated diffusion processes directly to the latent representations. By encoding the mask into a latent representation before diffusion and decoding it back only at the final step, we minimize the loss of detail and improve the overall fidelity of the reconstructed mask.

III. Experiments

A. Experiment Settings

1) Datasets: We conduct experiments on three widely-used COD datasets: CAMO [42], COD10K [1], and NC4K [32]. The CAMO dataset comprises 1,250 camouflaged images and 1,250 non-camouflaged images. COD10 K contains 5,066 camouflaged images, 3,000 background images, and 1,934 non-camouflaged images. NC4K is a large-scale COD dataset consisting of 4,121 images and is often used to evaluate the generalizability of models. In addition to the COD datasets, we have also conducted extended experiments on the Salient Object Detection (SOD) datasets. For these experiments, we use commonly used benchmark datasets, including DUT-OMRON [43], which contains 5,168 images, DUTS-TE [44] with 5,019 images, and HKU-IS [45], which includes 4,447 images.

2) Evaluation Metrics: In our experiments, we adopt four commonly used metrics:

Mean Absolute Error (MAE) [46] quantifies the average absolute error between the normalized predicted map and the ground truth.

S-measure (S_α) [47] evaluates the structural similarity between the predicted map and the ground truth map, considering both object-aware and region-aware features.

Weighted F-measure (F_β^ω) [48] provides a comprehensive evaluation of precision and recall, effectively measuring the accuracy of the predicted map.

评估指标：平均绝对误差，s测量，F加权度量，E平均测量值

Mean E-measure (E_ϕ) [49] is based on human visual perception and provides an assessment of the overall and local accuracy of camouflaged object detection results.

3) Implementation Details: We conduct all experiments on PyTorch platform by using an NVIDIA A100 GPU. The ATCN utilizes PVTv2-B4 for initialization, processing images resized to 384×384. We applied AdamW [50] as the optimizer, setting the batch size to 32. The learning rate follows a cosine adjustment strategy, starting at 0.001 across 170 epochs. Given the inherent variability in our sampling approach, the evaluations are performed three times using different seeds, and the metric averages are reported. CamoDiffusion-E, an extension, employs the CTE strategy for triple inference passes to increase reliability. The standard setup utilizes $T = 10$ for the sampling process.

B. Quantitative and Qualitative Evaluation

1) COD Task Evaluation: Table I details the quantitative achievements of our model, contrasting it with 14 competitive entries across a trio of COD benchmark datasets. We present three versions of our method: CamoDiffusion-VAE, CamoDiffusion and CamoDiffusion-E. CamoDiffusion-VAE employs VQ-VAE to accelerate the process with reduced computational complexity, as detailed in Section II-E. CamoDiffusion-E represents the inference stage using the CTE strategy. Remarkably, all three versions of our model achieve state-of-the-art performance, even the lightweight CamoDiffusion-VAE.

Specifically, our model is superior to other methods compared by significantly reducing the MAE error by a notable 20.9% and amplifying the F_β^w score by 7.7%, surpassing the second-best method, i.e., FSPNet [38]. The essence of this unparalleled performance lies in the iterative denoising prowess inherent to diffusion models, which is further refined by the application of our CTE strategy. Unique among its diffusion model counterparts, our model introduces specific enhancements, meticulously designed to address the intricate demands of the COD discipline.

Fig. 7 provides a graphical comparison between CamoDiffusion and several leading models, illustrating the qualitative distinctions. The initial segments, represented by the first six rows, cover a spectrum of common situations encountered in

TABLE I
QUANTITATIVE RESULTS OF OUR METHOD AND OTHER STATE-OF-THE-ART METHODS ON THREE COD BENCHMARK DATASETS

Baseline Models	CAMO(250)				COD10K(2026)				NC4K(4121)			
	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^w \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^w \uparrow$	$M \downarrow$	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^w \uparrow$	$M \downarrow$
EGNet [26]	0.662	0.683	0.495	0.125	0.733	0.761	0.519	0.055	0.767	0.793	0.626	0.077
F3Net [27]	0.711	0.741	0.564	0.109	0.739	0.795	0.544	0.051	0.780	0.824	0.656	0.070
SINet [28]	0.745	0.804	0.644	0.092	0.776	0.864	0.631	0.043	0.808	0.871	0.723	0.058
MGL [29]	0.775	0.812	0.673	0.088	0.814	0.852	0.666	0.035	0.833	0.867	0.740	0.052
PFNet [30]	0.782	0.841	0.695	0.085	0.800	0.877	0.660	0.040	0.829	0.887	0.745	0.053
UGTR [31]	0.785	0.823	0.686	0.086	0.818	0.853	0.667	0.035	0.839	0.874	0.747	0.052
LSR [32]	0.787	0.838	0.696	0.080	0.804	0.880	0.673	0.037	0.840	0.895	0.766	0.048
PreyNet [33]	0.790	0.842	0.708	0.077	0.813	0.881	0.697	0.034	0.834	0.887	0.763	0.050
SINet-V2 [1]	0.820	0.882	0.743	0.070	0.815	0.887	0.680	0.037	0.847	0.903	0.770	0.048
ZoomNet [6]	0.820	0.877	0.752	0.066	0.838	0.888	0.729	0.029	0.853	0.896	0.784	0.043
DTINet [34]	0.856	0.916	0.796	0.050	0.824	0.896	0.695	0.034	0.863	0.917	0.792	0.041
FEDER-R50 [35]	0.802	0.867	0.738	0.071	0.822	0.900	0.716	0.032	0.847	0.907	0.789	0.044
DGNet [36]	0.839	0.901	0.769	0.057	0.822	0.896	0.693	0.033	0.857	0.911	0.784	0.042
HitNet [37]	0.849	0.906	0.809	0.055	0.871	0.935	0.806	0.023	0.875	0.926	0.834	0.037
FSPNet [38]	0.856	0.899	0.799	0.050	0.851	0.895	0.735	0.026	0.879	0.915	0.816	0.035
Diffusion Models for Image Segmentation												
EnsemDiff [39]	0.716	0.751	0.568	0.113	0.712	0.748	0.508	0.069	0.758	0.791	0.601	0.089
MedSegDiff [40]	0.735	0.801	0.641	0.098	0.731	0.812	0.548	0.047	0.785	0.824	0.664	0.069
PrObeD [41]	0.797	0.871	0.702	0.071	0.803	0.869	0.661	0.037	0.838	0.900	0.755	0.049
CamoDiffusion-VAE(Ours)	0.868	0.926	0.827	0.047	0.857	0.919	0.759	0.024	0.877	0.926	0.821	0.034
CamoDiffusion(Ours)	**0.878**	**0.940**	**0.853**	**0.042**	0.881	**0.944**	0.814	0.020	0.893	**0.942**	0.859	0.029
CamoDiffusion-E(Ours)	**0.878**	0.936	0.851	**0.042**	**0.883**	0.943	**0.817**	**0.019**	**0.895**	**0.942**	**0.861**	**0.028**

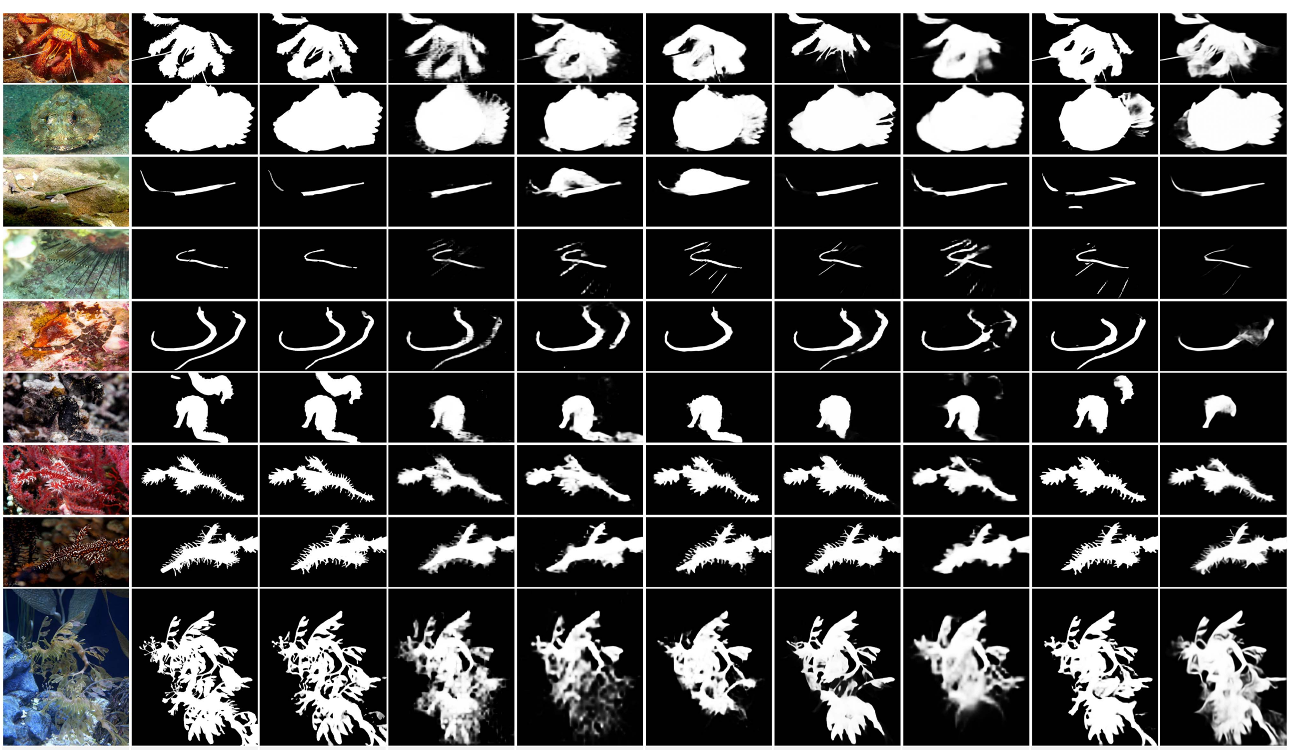


Fig. 7. Visual comparisons against recent SOTA models under typical and challenging conditions within the COD task.

TABLE II
QUANTITATIVE RESULTS OF OUR METHOD AND OTHER STATE-OF-THE-ART METHODS ON THREE SOD BENCHMARK DATASETS

Baseline Models	DUT-O				DUTS-TE				HKU-IS			
	$M\downarrow$	$E_\phi^m\uparrow$	$S_\alpha\uparrow$	$F_\beta^w\uparrow$	$M\downarrow$	$E_\phi^m\uparrow$	$S_\alpha\uparrow$	$F_\beta^w\uparrow$	$M\downarrow$	$E_\phi^m\uparrow$	$S_\alpha\uparrow$	$F_\beta^w\uparrow$
PiCANet [53]	0.054	0.865	0.826	0.743	0.040	0.915	0.863	0.812	0.031	0.951	0.905	0.89
BASNet [54]	0.056	0.871	0.836	0.751	0.048	0.903	0.866	0.803	0.032	0.951	0.909	0.889
CPD-R [55]	0.056	0.868	0.825	0.719	0.043	0.914	0.869	0.795	0.034	0.95	0.905	0.875
PoolNet [56]	0.054	0.867	0.831	0.725	0.037	0.926	0.887	0.817	0.030	0.958	0.919	0.888
AFNet [57]	0.057	0.861	0.826	0.717	0.046	0.910	0.867	0.785	0.036	0.949	0.905	0.869
EGNet [26]	0.053	0.878	0.841	0.738	0.039	0.927	0.887	0.816	0.031	0.958	0.918	0.887
ITSD-R [58]	0.061	0.880	0.840	0.750	0.041	0.929	0.885	0.824	0.031	0.960	0.917	0.894
MINet-R [59]	0.056	0.869	0.833	0.738	0.037	0.927	0.884	0.825	0.029	0.960	0.919	0.897
LDF [60]	0.052	0.869	0.839	0.752	0.034	0.930	0.892	0.845	0.028	0.958	0.919	0.904
CSF-R2 [61]	0.055	0.775	0.838	0.733	0.037	0.930	0.890	0.823	0.030	0.960	0.921	0.891
GateNet-R [62]	0.055	0.876	0.838	0.729	0.040	0.928	0.885	0.809	0.033	0.955	0.915	0.880
PFSNet [63]	0.055	0.878	0.842	0.756	0.036	0.931	0.892	0.842	0.026	0.962	0.924	0.910
VST [64]	0.058	0.890	0.852	0.758	0.037	0.940	0.897	0.831	0.029	0.968	0.928	0.897
SRformer [65]	0.043	0.894	0.860	0.784	0.027	0.952	0.910	0.872	0.025	0.966	0.928	0.912
CamoDiffusion	**0.042**	**0.903**	**0.871**	**0.806**	**0.023**	**0.958**	**0.921**	**0.902**	**0.022**	**0.972**	**0.934**	**0.929**

COD tasks. These include scenes populated with objects of diverse sizes or numerous entities, where CamoDiffusion notably minimizes the instances of overconfident mis-segmentation, showcasing its precision and reliability. The subsequent segments, rows 7 through 9, dive into more complex scenarios characterized by intricate topological structures and sharp edges. These scenarios pose substantial challenges to existing COD methodologies. However, our model demonstrates a clear competence to accurately define the boundaries in such demanding cases, leveraging the robust iterative denoising framework. This enhanced delineation ability stems from our model's innovative modifications, which are strategically crafted to optimize performance across a wide array of COD scenarios.

2) SOD Task Evaluation: Although our method primarily targets the COD task, prior studies have highlighted the connection between COD and Salient Object Detection (SOD) [28], [51], [52]. SOD and COD are related binary segmentation tasks—SOD identifies visually distinct objects, while COD targets concealed objects. Both tasks require effective boundary detection, segmentation, and multi-scale feature extraction, forming a natural basis for leveraging a unified model for both.

Our CamoDiffusion model capitalizes on these aspects through a conditional diffusion framework to progressively refine boundaries and reduce over-confident predictions. Previous works [9], [10] show that multi-scale integration benefits both tasks, albeit with different focuses: sharp edges in SOD versus subtle textures in COD. Given these similarities, extending our model to SOD was a natural step.

To demonstrate the versatility of our approach, we have extended our experiments to include the SOD task.

Table II provides a thorough quantitative comparison of our proposed method against 14 state-of-the-art models on three SOD benchmark datasets. The results unequivocally indicate that our method outshines all other contemporary state-of-the-art techniques, affirming its efficacy and superiority in meeting the demands of the task.

TABLE III
ABLATION OF MODEL COMPONENTS AND TRAINING STRATEGIES ON COD10 K DATASET

	ZOE	TTC	SNR	SC	$S_\alpha\uparrow$	$E_\phi\uparrow$	$F_\beta^w\uparrow$	$M\downarrow$
ATCN	×	×	×	×	0.865	0.927	0.774	0.024
Model Components	×	×	✓		0.867	0.933	0.783	0.022
	×	✓			0.872	0.938	0.795	0.022
	✓	×			0.877	0.940	0.806	**0.020**
	✓	✓			**0.881**	**0.944**	**0.814**	**0.020**
Training Strategies	✓		×	×	0.821	0.898	0.713	0.033
			✓	×	0.868	0.931	0.789	0.022
			×	✓	0.855	0.922	0.767	0.024
			✓	✓	**0.881**	**0.944**	**0.814**	**0.020**

All these results reveal that our method can capture the global visual context and the nuances of the saliency objects, providing a more reliable solution for both SOD and COD.

C. Ablation Studies

We conduct an ablation study on CamoDiffusion, as detailed in Tables III and IV, Figs. 8 and 9.

1) Ablation of Model Components: In the Architecture Design section, we introduced two key advancements: ZOE and TTC modules. To assess the significance of these components, We conduct an ablation study on the individual components of CamoDiffusion, with detailed results provided in Table III. Instances where both ZOE and TTC are marked with a '×' indicate the absence of ATCN, instead relying on a denoising process that utilizes consistent image features. The analysis reveals that our model achieves its highest performance when both ZOE and TTC are implemented. In contrast, removing ZOE results in a performance decrement, underscoring its vital role in effectively integrating mask information without altering the original image data. This finding highlights that ZOE contribute to maintain the integrity of the image while seamlessly blending in the mask features, thereby optimizing the overall performance.

TABLE IV
ABLATION OF OUR CTE STRATEGY ON COD10K

Sampling Strategy	$S_\alpha \uparrow$	$E_\phi \uparrow$	$F_\beta^w \uparrow$	$M \downarrow$	Overconfident False Positives	Overconfident False Negatives	Overconfident Incorrect Pixels	Total Incorrect Pixels	Percentage
w/o CTE	0.877	0.941	0.803	0.021	5185	4234	9419	10394	91%
CamoDiffusion	0.881	**0.944**	0.814	0.020	3582	4062	7644	8957	85%
CamoDiffusion-E	**0.883**	0.943	**0.817**	**0.019**	**3007**	**3642**	**6649**	**8698**	**76%**

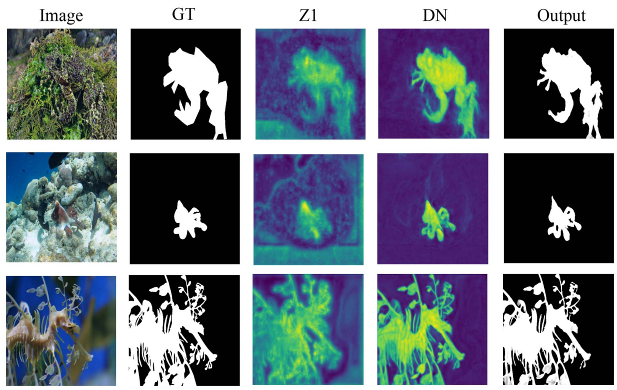


Fig. 8. Feature visualization of the of Z1 and penultimate layer of DN.

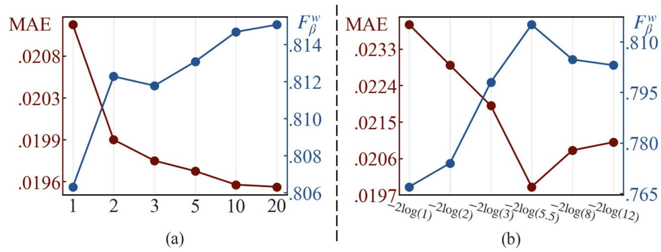


Fig. 9. Performance on different sample steps (a) and SNR shift (b).

2) Ablation of ATCN and Denoising Network (DN): The Adaptive Transformer Conditional Network (ATCN) and the Denoising Network (DN) play complementary roles in our camouflaged object detection framework. ATCN is designed to extract contextual features (denoted as Z_1) that serve as a coarse guide for the subsequent denoising process, while DN progressively refines these features to produce accurate segmentation masks.

Quantitative experiments show that relying solely on ATCN, without the subsequent refinement steps by DN, leads to a significant drop in performance. As summarized in Table III, the ATCN module alone struggles with capturing the finer segmentation details, resulting in lower metrics such as F_β^w and E_ϕ. This highlights the need for the iterative refinement provided by DN to achieve precise boundary delineation and robust object localization.

From a qualitative perspective, visualizations of Z_1 and the intermediate features within DN, as shown in Fig. 8, indicate that Z_1 provides an initial approximation of the target region, capturing the overall shape but with limitations like blurred boundaries and over-confident predictions. The DN, through the iterative diffusion steps, effectively refines these boundaries and reduces over-segmentation errors, resulting in a final mask with much greater accuracy and consistency. These analyses demonstrate that ATCN and DN work synergistically: ATCN supplies the coarse localization information, and DN iteratively corrects and refines it to produce a high-quality output.

3) Ablation of Training Strategies: The latter part of Table III showcases the ablation studies conducted for these strategies, demonstrating their individual and combined effectiveness. The adoption of the SNR strategy alone has led to a notable improvement in S-measure, from 0.821 to 0.868, validating the efficacy of our variation noise schedule. Additionally, our findings indicate that Structure Corruption also enhances performance, validating the effectiveness of this perturbation strategy for camouflaged targets. The combination of both strategies significantly increases performance on the COD10-K dataset, underlining the synergistic benefit of integrating the SNR-based variation schedule with structure corruption.

4) Ablation of Sampling Strategy: Table IV offers a comprehensive analysis of the ablation study for the CTE strategy, detailing pixel counts for both over-segmentation and under-segmentation. The results highlight a significant decrease in overconfident false predictions, particularly with over-segmentation (overconfident false positives). This improvement stems from the CTE mechanism's ability to disregard areas where less than half of the segmentation maps yield incorrect predictions. In scenarios where the majority of segmentation maps produce inaccurate predictions, the CTE strategy incorporates correct predictions to diminish confidence in these areas. Consequently, the proportion of overconfident incorrect pixels fell from 91% to 76%, marking a substantial increase in the model's reliability. This strategic approach underlines the CTE strategy's effectiveness in enhancing segmentation accuracy and model dependability.

5) Ablation of Hyperparameters: Fig. 9(a) shows how the effectiveness of the model in the COD10 k dataset varies with the number of sampling steps. An increase in steps correlates with improved performance, though it also leads to extended inference durations. For optimal results, setting the sampling to 10 steps is recommended, whereas a setting of 3 steps offers a balance between performance and efficiency in inference times.

Moreover, Fig. 9(b) examines the effect of the SNR shift on the efficacy of the model. In particular, lowering the shift value reduces the amount of mask information retained in $\mathbf{x}_t$, which in turn increases the complexity of training. The model achieves peak performance at a shift value of $-2\log(5.5)$, indicating the importance of carefully adjusting the SNR shift.

TABLE V
COMPARISON OF THE COMPLEXITY OF SEVERAL EXISTING METHODS ON THE COD10 K DATASET IN TERMS OF PARAMETERS, MACS AND FPS

Models	Publications	Input	Backbone	Steps.	Para.	MACs	FPS	CAMO $M\downarrow$	COD10K $M\downarrow$	NC4K $M\downarrow$
MGL [29]	CVPR2021	473^2	ResNet-50	N/A	67.64M	236.6G	28.84	0.088	0.035	0.052
UGTR [31]	ICCV2021	473^2	ResNet-50	N/A	48.87M	121.7G	28.81	0.086	0.035	0.052
ZoomNet [6]	CVPR2022	384^2	ResNet-50	N/A	32.38M	88.97G	36.07	0.066	0.029	0.043
DTINet [34]	ICPR2022	256^2	Mit-B5	N/A	266.33M	144.7G	11.74	0.050	0.034	0.041
HitNet [37]	AAAI2023	702^2	PVTv2-B2	N/A	25.73M	56.55G	42.70	0.055	0.023	0.037
FSPNet [38]	CVPR2023	384^2	ViT-B	N/A	274.24M	283.3G	35.56	0.050	0.026	0.035
①: CamoDiffusion	-	384^2	PVTv2-B4	10	71.70M	60.29G/S	28.80/S	0.042	0.020	0.029
②: ① + reduce channel	-	384^2	PVTv2-B4	10	68.73M	45.58G/S	28.80/S	**0.043**	**0.020**	**0.029**
③: ② + lower resolution	-	352^2	PVTv2-B4	10	68.73M	38.31G/S	29.30/S	**0.043**	0.021	**0.029**
④: ① + smaller backbone	-	384^2	PVTv2-B2	10	32.75M	31.63G/S	58.90/S	0.048	0.022	0.032
⑤: ① + VQ-VAE	-	384^2	PVTv2-B4	10	72.00M	34.88G/S	30.53/S	0.046	0.024	0.034
⑥: ② + fewer steps	-	384^2	PVTv2-B4	3	68.73M	45.58G/S	28.80/S	0.044	0.021	**0.029**
⑦: ③ + fewer steps	-	352^2	PVTv2-B4	3	68.73M	38.31G/S	29.30/S	0.044	0.021	0.030
⑧: ④ + fewer steps	-	384^2	PVTv2-B2	3	32.75M	31.63G/S	58.90/S	0.050	0.022	0.033
⑨: ⑦ + lower resolution	-	352^2	PVTv2-B2	3	32.75M	26.58G/S	29.30/S	0.051	0.023	0.033
⑩: ④ + VQ-VAE	-	384^2	PVTv2-B2	3	36.00M	**17.77**G/S	74.53/S	0.050	0.025	0.034
❶: ③ + ATCN-Skip	-	352^2	PVTv2-B4	3	68.73M	30.17G/S	39.90/S	**0.043**	0.021	0.029
❷: ④ + ATCN-Skip	-	384^2	PVTv2-B2	3	32.75M	20.25G/S	**89.94**/S	0.052	0.023	0.032

We distinguish our model in the paper (marked as ①), and present variants. By reducing network's channels, input resolution and denoising steps, our model yields outstanding results while maintaining acceptable computational costs.

D. Analysis

1) Analysis of Acceleration Strategy: Incorporating diffusion models into our framework aims to tackle two fundamental challenges in the COD task, resulting in a significant improvement in performance. However, the intrinsic characteristics of diffusion models introduce potential drawbacks, including heightened computational demands. In this section, we examine various optimization strategies to mitigate these concerns.

In Table V, we investigate various configurations within our foundational model (①). The Variant ② realizes a significant reduction in computational costs by decreasing the denoising network channels from 256 to 192, but preserves the model performance. Furthermore, the Variant ③ computation by scaling down the input network's resolution from 384 to 352, which had a negligible effect on performance. To further improve efficiency, the Variant ④ switched to a lighter backbone (PVTv2-B2). By reducing the number of sampling steps during testing from 10 to 3, we effectively eased the computational load while still achieving commendable segmentation results.

Furthermore, as detailed in Section II-E, we evaluate the combination of VQ-VAE with two backbones, i.e., PVTv2-B4 (Variant ⑤) and PVTv2-B2 Variant ⑩, to further reduce computational requirements.

We can observe that among the strategies explored, ⑨ stands out as the optimal acceleration approach. It incurs a total computational cost of 79.74 G, which is marginally lower than ZoomNet's 88.97 G while achieving a substantial 21.6% reduction in average MAE. Compared to the recently proposed HitNet, our model shows a 6. 0% improvement in average MAE.

Furthermore, compared to FSPNet, our method reduces MAE by 5.1%, along with notable decreases in model parameters and computational expenses. This highlights the efficiency and effectiveness of our approach in optimizing performance while managing computational resources.

We also applied the ATCN-Skip strategy with both PVT-B4 and PVT-B2 as backbones. Compared to the original version, employing the ATCN-Skip strategy shows negligible accuracy drop while significantly improving inference speed. For instance, using PVT-B4 as the backbone, the MAE on the CAMO dataset remains the same at 0.043, and the COD10 K dataset sees only a slight increase from 0.020 to 0.021. On the NC4K dataset, the MAE remains consistent at 0.029. However, the FPS improves significantly from 28.8 to 39.9, representing a 38.5% increase in inference speed.

TABLE VI
COMPARISON OF MODELS THAT SHARE THE SAME BACKBONE (PVTV2-B2) ON THREE DATASETS

Model	CAMO $M\downarrow$	COD10K $M\downarrow$	NC4K $M\downarrow$	Avg. Imp.
Baseline	0.057	0.031	0.042	-
DGNet* [36]	0.051	0.029	0.037	10.64%
HitNet [37]	0.055	0.023	0.037	14.83%
CamoDiffusion*	**0.048**	**0.022**	**0.032**	23.83%

'*' means which utilizing backbones distinct from those in Table 1

Similarly, when using PVT-B2 as the backbone, the ATCN-Skip strategy shows a small increase in MAE on the CAMO dataset (from 0.048 to 0.052) and on COD10 K (from 0.022 to 0.023), while the FPS increased from 58.9 to 89.9. These results indicate that while there may be slight performance degradation on certain metrics, the increase in inference efficiency is substantial, making this strategy highly practical for real-time applications where speed is crucial.

Table V reveals several key insights. First, most acceleration strategies not only speed up the process, but also surpass the performance of traditional state-of-the-art (SOTA) methods. Second, our approach offers considerable scope for acceleration. Certain strategies do not significantly compromise performance, but substantially reduce overall computational demands, such as employing fewer steps.

2) Analysis of Backbone Influence: To better understand the influence of the diffusion process, we evaluate our method against two well-known COD approaches while controlling for the consistent backbone architecture. As shown in Table VI,

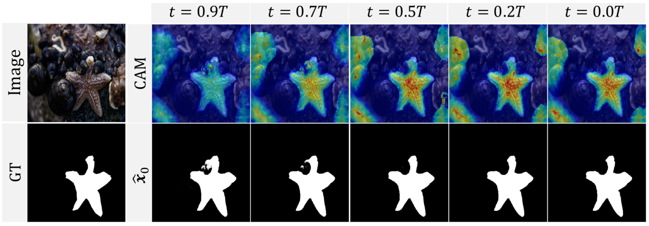


Fig. 10. The mask predictions and feature maps at different sample steps.

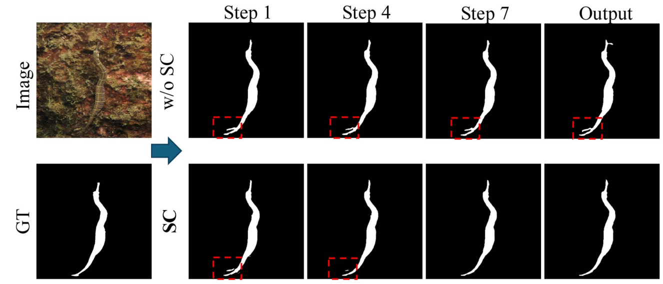


Fig. 11. Visualization of CamoDiffusion with and without the Structure Corruption (SC) module. The red dashed boxes highlight areas with segmentation errors.

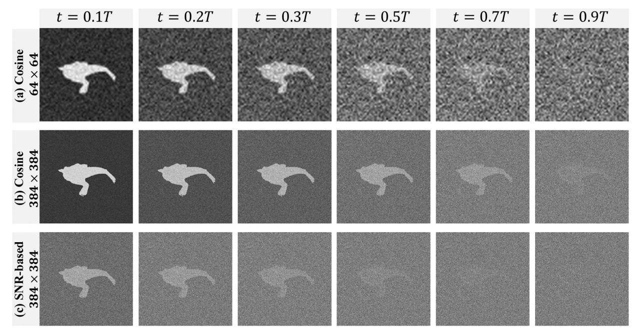


Fig. 12. The comparison of the consine variance schedule and the SNR-based variance schedule at variaous time steps and mask sizes. The first two rows demonstrate that the consine variance schedule is inappropriate for high-resolution binary masks in COD, as the mask is still visible at a resolution of 384 even when $t = 0.9$ T. The last two rows demonstrate how, at the same time step, using SNR-based variance schedule results in a smaller signal-to-noise ratio, enabling the model to learn and recover clear masks from image features more effectively.

we compare our method with DGNet, HitNet, and a baseline model, all of which employ the same PVTv2-B2 backbone as ours. The "Baseline" model, which follows the traditional segmentation approach, incorporates the PVTv2-B2 backbone and several convolutional layers. DGNet* is explored using its publicly available official code.

Our method outperforms these models, achieving a 23.83% improvement over the baseline. This performance gain, observed with a consistent backbone across methods, highlights the importance of the diffusion model in advancing the COD task. By isolating the impact of the backbone, these results demonstrate the effectiveness of our approach and the significance of the diffusion process in enhancing COD performance.

3) Analysis of Sampling Stages: To delve deeper into the diffusion model's step-by-step denoising process, we visualized the mask results and corresponding Class Activation Maps (CAM) at each step, as shown in Fig. 10. This visualization illustrates our model's adeptness at progressively reducing noise and focusing on detailed aspects through the depiction of prediction outcomes across different sampling stages.

Our model produces a basic mask at $0.9T$, marked by considerable uncertainty in regions where borders are not well defined. With the increment in sampling steps, there is a notable improvement in the model's precision in isolating and refining the mask surrounding the camouflaged object, thereby delineating clear boundaries that reflect the intricate details of the foreground. This stepwise refinement process is made possible through the deliberate incorporation of diffusion models, which circumvents the need for complex refinement mechanisms in existing COD models. The visual representations highlight the model's proficiency in progressively enhancing detail and accuracy via iterative enhancements, demonstrating the potency of diffusion models in improving segmentation results within the realm of COD.

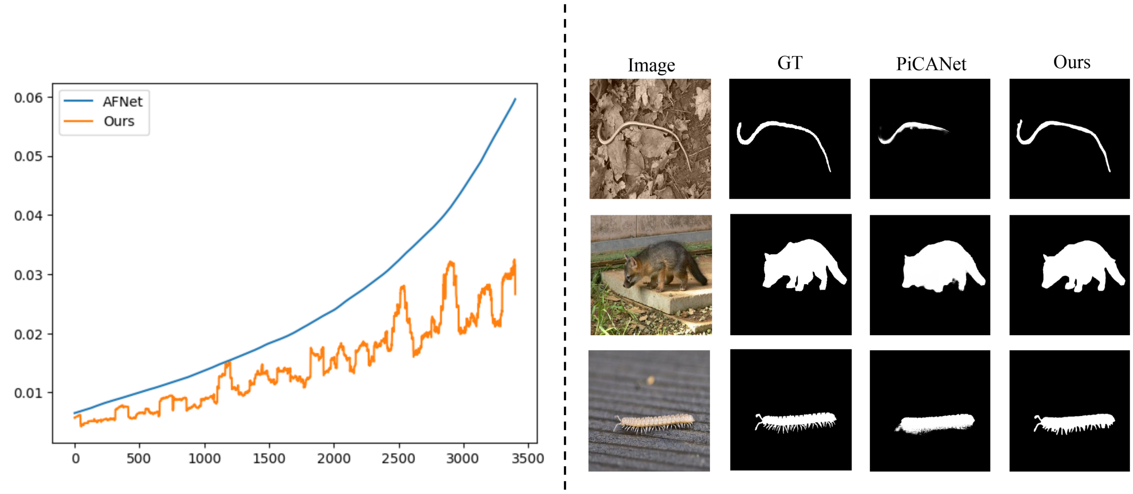
4) Analysis of Structure Corruption: We qualitatively analyze the ability of the SC strategy, so that we visualize the results of each step with or without SC. We show the results in Fig. 11. Here red box highlights the area with significant segmentation errors.

We observe that the model without the SC module lacks the ability to correct segmentation errors, although both models incorrectly segment the regions located in the red boxes in the initial stages (such as step 1). In contrast, the model with the SC module introduces additional corruption to x_t, allowing the network to recognize that x_t may contain errors. Consequently, the model gradually refines the mask, ultimately producing a more accurate and detailed output.

The SC strategy not only refines the edges but also addresses the overconfidence problem by enabling the network to progressively eliminate erroneous confidences. By corrupting the structure of the noised mask, the SC module forces the model to rely more on the visual features of the input image rather than solely on the noised mask's contours. This encourages the model to learn to make corrections based on the relevant visual cues, leading to improved segmentation accuracy and robustness, especially in challenging scenarios involving camouflaged objects.

5) Analysis of SNR-Based Variance Schedule: To evaluate the effectiveness of different variance schedules in corrupting binary masks across various resolutions, we conduct an experiment illustrated in Fig. 12. The results show that the cosine variance schedule faces challenges when corrupting binary masks with a resolution of 384 × 384. Specifically, masks with a resolution of 384 remain recognizable even at $t = 0.7T$, while masks with a resolution of 64 undergo substantial distortion. This indicates that the cosine variance schedule may not be optimal for handling high-resolution masks in COD tasks.

In contrast, the SNR-based variance schedule demonstrates performance on par with the cosine variance schedule during



(a). MAE comparison on DUT-STE **(b). SOD output visualization**

Fig. 13. (a) MAE comparison of AFNet and our method on DUT-STE dataset samples, ordered by increasing difficulty. (b) Qualitative results of SOD outputs for selected challenging samples, comparing PiCANet, ground truth, and our method.

the initial corruption process for masks of lower resolution. This suggests that the SNR-based schedule is effective across different stages of mask corruption and can adapt to the unique challenges posed by high-resolution masks. By considering the signal-to-noise ratio, the SNR-based schedule provides a more robust and efficient approach to corrupting binary masks in COD tasks, especially when dealing with high-resolution images.

6) Unified Analysis of COD and SOD Tasks: To better understand the gains provided by our unified approach, we conducted additional analyses of SOD performance, specifically targeting difficult samples. Quantitative analysis, as illustrated in Fig. 13(a), shows the Mean Absolute Error (MAE) on samples from the DUT-STE dataset, sorted in increasing difficulty. While the performance difference between our model and AFNet is minor for easier samples, the gap becomes significantly larger for challenging samples, indicating that our model excels particularly on complex cases. This is further supported by qualitative results in Fig. 13(b), which show that our model outperforms baseline methods on difficult SOD samples, especially in identifying objects with camouflaged characteristics and refining their boundaries effectively.

These results indicate that the techniques developed for COD, such as handling blurred boundaries and deceptive textures, also enhance SOD performance, particularly for challenging scenes. The shared challenges of ambiguous boundaries and over-confident predictions in both COD and SOD make our unified framework feasible and beneficial, achieving strong results on both tasks through effective cross-task knowledge transfer and shared model components.

IV. Related Work

A. Camouflaged Object Detection

Recent years have seen notable advancements in Camouflaged Object Detection (COD), broadly categorized into three

多流框架：不同输入源的全面场景检测
分层特征增强：自下而上，自上而下，丰富特征
多样化任务框架：多功能网络，同时处理任务

strategies: (1) *Multi-stream framework:* Leveraging diverse input sources for a comprehensive scene representation, enabling nuanced detection of camouflaged objects [6]. (2) *Hierarchical Feature Enhancement:* Using bottom-up and top-down approaches to enrich features by integrating semantic and surface-level details, progressively improving object perception [1]. (3) *Diversified Task Architecture:* Employing versatile networks that tackle segmentation alongside auxiliary tasks, offering a holistic approach to COD [36].

Prominent models like SINet-V2 [1] and BASNet [54] use multi-stage refinement for segmentation. Multi-task learning approaches, such as LSR [32] and DGNet [36], combine tasks like ranking and localization to improve outcomes. ZoomNet [6] and MFFN [66] enhance feature extraction through augmented inputs. Vision Transformer-based methods such as DTINet [34] and HitNet [37] employ attention mechanisms for long-range dependencies, showing promising COD performance. Despite advancements, COD models still face challenges like overconfident mis-segmentation and high architectural complexity.

B. Salient Object Detection

Recent advancements in Salient Object Detection (SOD) have been largely driven by CNN-based methods. Multi-task learning strategies have also been employed, as seen in EGNet [26] and CPD [55], which integrate edge detection with saliency prediction to enhance accuracy. Innovations in instance-level attention and gating mechanisms are exemplified by GateNet [62], offering improved performance in complex scenes. Furthermore, Vision Transformer-based methods such as Trans2Seg [67] and VST [64] have shown promising results by leveraging self-attention mechanisms for long-range dependency modeling. Despite these advancements, challenges persist in overconfident mis-segmentation and the complexity of model architectures, underscoring the need for continued research and optimization in the field of SOD. SOD and COD share many underlying challenges, such as handling ambiguous boundaries and overconfident predictions [9], [10]. Both tasks require accurate boundary detection, multi-scale feature integration, and robust differentiation of target objects from the background. Our work builds on these commonalities to develop a unified framework capable of handling both tasks effectively, demonstrating that the techniques used to address COD can also lead to significant improvements in the SOD domain.

C. Diffusion Models for Image Segmentation

Diffusion models [11], [23], [24] are structured as parameterized Markov chains, initiating the denoising of data samples from pure noise. Initially used in domains lacking clear ground truths [19], they have since proved their worth in areas with distinct solutions, like super-resolution [68], [69], deblurring [70], [71], and image segmentation [16], [17], [72]. This study underscores diffusion models' versatility across segmentation applications, including remote sensing change detection [73] and medical image segmentation [39], [40], [74]. Despite their potential, diffusion-based techniques remain a step behind in general image segmentation tasks, especially for COD challenges. Here, traditional diffusion methods often produce features with insufficient discriminatory capability and struggle with mask refinement.

V. Conclusion

In this paper, we introduce **CamoDiffusion**, a novel approach that tackles the COD and SOD tasks through the lens of conditional diffusion models. CamoDiffusion addresses two key challenges in COD and SOD: ambiguous boundaries and overconfident predictions. Our method generates the mask by progressive denoising a noise map under a conditional context obtained from the camouflaged image. To address the distinct challenges posed by COD and SOD, we propose a comprehensive framework that integrates architectural innovations, novel training techniques, and optimized sampling strategies. Our approach is specifically designed to capture the subtle details and boundaries of camouflaged objects while mitigating the issue of overconfident predictions. Experiments show that CamoDiffusion achieves state-of-the-art performance on three COD datasets and several SOD datasets, demonstrating its effectiveness and versatility. Our method's ability to address issues of over-segmentation and mis-segmentation in both COD and SOD tasks highlights its robustness. In future work, we plan to explore the application of our conditional diffusion framework to a wider range of segmentation tasks and investigate the integration of more advanced diffusion models to further improve performance and expand its applicability.

References

[1] D.-P. Fan, G.-P. Ji, M.-M. Cheng, and L. Shao, "Concealed object detection," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 10, pp. 6024–6042, Oct. 2022.

[2] D.-P. Fan, G.-P. Ji, P. Xu, M.-M. Cheng, C. Sakaridis, and L. Van Gool, "Advances in deep concealed scene understanding," *Vis. Intell.*, vol. 1, no. 1, 2023, Art. no. 16.

[3] M. G. Nafus, J. M. Germano, J. A. Perry, B. D. Todd, A. Walsh, and R. R. Swaisgood, "Hiding in plain sight: A study on camouflage and habitat selection in a slow-moving desert herbivore," *Behav. Ecol.*, vol. 26, no. 5, pp. 1389–1394, 2015.

[4] B. Dong, W. Wang, D.-P. Fan, J. Li, H. Fu, and L. Shao, "Polyp-PVT: Polyp segmentation with pyramid vision transformers," 2021, *arXiv:2108.06932*.

[5] N. U. Bhajantri and P. Nagabhushan, "Camouflage defect identification: A novel approach," in *Proc. IEEE 9th Int. Conf. Inf. Technol.*, 2006, pp. 145–148.

[6] Y. Pang, X. Zhao, T.-Z. Xiang, L. Zhang, and H. Lu, "Zoom in and out: A mixed-scale triplet network for camouflaged object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 2150–2160.

[7] J. Zhang et al., "UC-Net: Uncertainty inspired RGB-D saliency detection via conditional variational autoencoders," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 8579–8588.

[8] Y. Zhang, J. Zhang, W. Hamidouche, and O. Deforges, "Predictive uncertainty estimation for camouflaged object detection," *IEEE Trans. Image Process.*, vol. 32, pp. 3580–3591, 2023.

[9] J. Zhu, X. Zhang, S. Zhang, and J. Liu, "Inferring camouflaged objects by texture-aware interactive guidance network," in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2021, pp. 3599–3607.

[10] Z. Luo et al., "VSCode: General visual salient and camouflaged object detection with 2D prompt learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2024, pp. 17 169–17 180.

[11] J. Ho, A. Jain, and P. Abbeel, "Denoising diffusion probabilistic models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2020, pp. 6840–6851.

[12] R. Rombach, A. Blattmann, D. Lorenz, P. Esser, and B. Ommer, "High-resolution image synthesis with latent diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 10684–10695.

[13] Z. Chen, K. Sun, X. Lin, and R. Ji, "CamoDiffusion: Camouflaged object detection via conditional diffusion models," in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2024, pp. 1272–1280.

[14] W. Wang et al., "PVT v2: Improved baselines with pyramid vision transformer," *Comput. Vis. Media J.*, vol. 8, no. 3, pp. 415–424, 2022.

[15] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," 2016, *arXiv:1607.06450*.

[16] T. Amit, E. Nachmani, T. Shaharbany, and L. Wolf, "SegDiff: Image segmentation with diffusion probabilistic models," 2021, *arXiv:2112.00390*.

[17] T. Chen, L. Li, S. Saxena, G. Hinton, and D. J. Fleet, "A generalist framework for panoptic segmentation of images and videos," 2022, *arXiv:2210.06366*.

[18] W. Shi et al., "Real-time single image and video super-resolution using an efficient sub-pixel convolutional neural network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2016, pp. 1874–1883.

[19] P. Dhariwal and A. Nichol, "Diffusion models beat GANs on image synthesis," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 8780–8794.

[20] J. Wang, Q. Huang, F. Tang, J. Meng, J. Su, and S. Song, "Stepwise feature fusion: Local guides global," in *Proc. 25th Int. Conf. Med. Image Comput. Comput. Assist. Intervention*, Singapore, Springer, Sep. 18–22, 2022, pp. 110–120.

[21] E. Hoogeboom, J. Heek, and T. Salimans, "simple diffusion: End-to-end diffusion for high resolution images," 2023, *arXiv:2301.11093*.

[22] D. Kingma, T. Salimans, B. Poole, and J. Ho, "Variational diffusion models," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 21 696–21 707.

[23] Y. Song, J. Sohl-Dickstein, D. P. Kingma, A. Kumar, S. Ermon, and B. Poole, "Score-based generative modeling through stochastic differential equations," 2020, *arXiv: 2011.13456*.

[24] Y. Song and S. Ermon, "Generative modeling by estimating gradients of the data distribution," in *Proc. Adv. Neural Inf. Process. Syst.*, 2019, pp. 11918–11930.

[25] S. Gu et al., "Vector quantized diffusion model for text-to-image synthesis," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 10 696–10 706.

[26] J.-X. Zhao, J.-J. Liu, D.-P. Fan, Y. Cao, J. Yang, and M.-M. Cheng, "EGNet: Edge guidance network for salient object detection," in *Proc. Int. Conf. Comput. Vis.*, 2019, pp. 8779–8788.

[27] J. Wei, S. Wang, and Q. Huang, "F ^{3} Net: Fusion, feedback and focus for salient object detection," in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2020, pp. 12321–12328.

[28] D.-P. Fan, G.-P. Ji, G. Sun, M.-M. Cheng, J. Shen, and L. Shao, "Camouflaged object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 2777–2787.

[29] Q. Zhai, X. Li, F. Yang, C. Chen, H. Cheng, and D.-P. Fan, "Mutual graph learning for camouflaged object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 12997–13007.

[30] H. Mei, G.-P. Ji, Z. Wei, X. Yang, X. Wei, and D.-P. Fan, "Camouflaged object segmentation with distraction mining," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 8772–8781.

[31] F. Yang et al., "Uncertainty-guided transformer reasoning for camouflaged object detection," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 4146–4155.

[32] Y. Lv et al., "Simultaneously localize, segment and rank the camouflaged objects," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 11591–11601.

[33] M. Zhang, S. Xu, Y. Piao, D. Shi, S. Lin, and H. Lu, "PreyNet: Preying on camouflaged objects," in *Proc. ACM Int. Conf. Multimedia*, 2022, pp. 5323–5332.

[34] Z. Liu, Z. Zhang, Y. Tan, and W. Wu, "Boosting camouflaged object detection with dual-task interactive transformer," in *Proc. Int. Conf. Pattern Recognit.*, 2022, pp. 140–146.

[35] C. He et al., "Camouflaged object detection with feature decomposition and edge reconstruction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 22046–22055.

[36] G.-P. Ji, D.-P. Fan, Y.-C. Chou, D. Dai, A. Liniger, and L. Van Gool, "Deep gradient learning for efficient camouflaged object detection," *Mach. Intell. Res.*, vol. 20, pp. 92–108, 2023.

[37] X. Hu et al., "High-resolution iterative feedback network for camouflaged object detection," in *Proc. Conf. Assoc. Advance. Artif. Intell.*, 2023, pp. 881–889.

[38] Z. Huang et al., "Feature shrinkage pyramid for camouflaged object detection with transformers," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 5557–5566.

[39] J. Wolleb, R. Sandkühler, F. Bieder, P. Valmaggia, and P. C. Cattin, "Diffusion models for implicit image segmentation ensembles," in *Proc. Int. Conf. Med. Imag. Deep Learn.*, PMLR, 2022, pp. 1336–1348.

[40] J. Wu, H. Fang, Y. Zhang, Y. Yang, and Y. Xu, "MedSegDiff: Medical image segmentation with diffusion probabilistic model," 2022, *arXiv:2211.00611*.

[41] V. Asnani, A. Kumar, S. You, and X. Liu, "PrObeD: Proactive object detection wrapper," *Adv. Neural Inf. Process. Syst.*, 2024, pp. 77993–78005.

[42] T.-N. Le, T. V. Nguyen, Z. Nie, M.-T. Tran, and A. Sugimoto, "Anabranch network for camouflaged object segmentation," *Comput. Vis. Image Understanding*, vol. 184, pp. 45–56, 2019.

[43] C. Yang, L. Zhang, H. Lu, X. Ruan, and M.-H. Yang, "Saliency detection via graph-based manifold ranking," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2013, pp. 3166–3173.

[44] L. Wang et al., "Learning to detect salient objects with image-level supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2017, pp. 136–145.

[45] G. Li and Y. Yu, "Visual saliency based on multiscale deep features," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 5455–5463.

[46] F. Perazzi, P. Krähenbühl, Y. Pritch, and A. Hornung, "Saliency filters: Contrast based filtering for salient region detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2012, pp. 733–740.

[47] D.-P. Fan, M.-M. Cheng, Y. Liu, T. Li, and A. Borji, "Structure-measure: A new way to evaluate foreground maps," in *Proc. Int. Conf. Comput. Vis.*, 2017, pp. 4548–4557.

[48] R. Margolin, L. Zelnik-Manor, and A. Tal, "How to evaluate foreground maps?," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 248–255.

[49] D.-P. Fan, G.-P. Ji, X. Qin, and M.-M. Cheng, "Cognitive vision inspired object segmentation metric and loss function," *Scientia Sinica Informationis*, vol. 6, 2021, Art. no. 5.

[50] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. Int. Conf. Learn. Representations*, 2019, pp. 1–10.

[51] X. Jiang et al., "Camouflaged object segmentation based on joint salient object for contrastive learning," *IEEE Trans. Instrum. Meas.*, vol. 72, 2023, Art. no. 5023816.

[52] A. Li, J. Zhang, Y. Lv, B. Liu, T. Zhang, and Y. Dai, "Uncertainty-aware joint salient object and camouflaged object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2021, pp. 10066–10076.

[53] N. Liu, J. Han, and M.-H. Yang, "PiCANet: Learning pixel-wise contextual attention for saliency detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 3089–3098.

[54] X. Qin, Z. Zhang, C. Huang, C. Gao, M. Dehghan, and M. Jagersand, "BASNet: Boundary-aware salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 7479–7489.

[55] Z. Wu, L. Su, and Q. Huang, "Cascaded partial decoder for fast and accurate salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 3907–3916.

[56] J.-J. Liu, Q. Hou, M.-M. Cheng, J. Feng, and J. Jiang, "A simple pooling-based design for real-time salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 3917–3926.

[57] M. Feng, H. Lu, and E. Ding, "Attentive feedback network for boundary-aware salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 1623–1632.

[58] H. Zhou, X. Xie, J.-H. Lai, Z. Chen, and L. Yang, "Interactive two-stream decoder for accurate and fast saliency detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9141–9150.

[59] Y. Pang, X. Zhao, L. Zhang, and H. Lu, "Multi-scale interactive network for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 9413–9422.

[60] J. Wei, S. Wang, Z. Wu, C. Su, Q. Huang, and Q. Tian, "Label decoupling framework for salient object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 13 025–13 034.

[61] S.-H. Gao, Y.-Q. Tan, M.-M. Cheng, C. Lu, Y. Chen, and S. Yan, "Highly efficient salient object detection with 100k parameters," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 702–721.

[62] X. Zhao, Y. Pang, L. Zhang, H. Lu, and L. Zhang, "Suppress and balance: A simple gated network for salient object detection," in *Proc. 16th Eur. Conf. Comput. Vis.*, Glasgow, UK, Springer, Aug. 23–28, 2020, pp. 35–51.

[63] M. Ma, C. Xia, and J. Li, "Pyramidal feature shrinking for salient object detection," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 2311–2318.

[64] N. Liu, N. Zhang, K. Wan, L. Shao, and J. Han, "Visual saliency transformer," in *Proc. Int. Conf. Comput. Vis.*, 2021, pp. 4722–4732.

[65] Y. K. Yun and W. Lin, "Selfreformer: Self-refined network with transformer for salient object detection," 2022, *arXiv:2205.11283*.

[66] D. Zheng, X. Zheng, L. T. Yang, Y. Gao, C. Zhu, and Y. Ruan, "MFFN: Multi-view feature fusion network for camouflaged object detection," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2023, pp. 6232–6242.

[67] E. Xie et al., "Segmenting transparent object in the wild with transformer," 2021, *arXiv:2101.08461*.

[68] H. Li et al., "SRDiff: Single image super-resolution with diffusion probabilistic models," *Neurocomputing*, vol. 479, pp. 47–59, 2022.

[69] C. Wang, K. Yeo, X. Jin, A. C. Duarte, L. Klein, and B. Elmegreen, "S3RP: Self-supervised super-resolution and prediction for advection-diffusion process," in *Proc. Annu. Conf. Neural Inf. Process. Syst.*, 2021, pp. 1–5.

[70] J. Whang, M. Delbracio, H. Talebi, C. Saharia, A. G. Dimakis, and P. Milanfar, "Deblurring via stochastic refinement," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 16 293–16 303.

[71] S. Lee, H. Chung, J. Kim, and J. C. Ye, "Progressive deblurring of diffusion models for coarse-to-fine image synthesis," in *Proc. NeurIPS 2022 Workshop Score-Based Methods*, 2022, pp. 1–6.

[72] D. Baranchuk, A. Voynov, I. Rubachev, V. Khrulkov, and A. Babenko, "Label-efficient semantic segmentation with diffusion models," in *Proc. Int. Conf. Learn. Representations*, 2021, pp. 1–11.

[73] W. G. C. Bandara, N. G. Nair, and V. M. Patel, "DDPM-CD: Remote sensing change detection using denoising diffusion probabilistic models," 2022, *arXiv:2206.11892*.

[74] A. Rahman, J. M. J. Valanarasu, I. Hacihaliloglu, and V. M. Patel, "Ambiguous medical image segmentation using diffusion models," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2023, pp. 11536–11546.

Ke Sun is currently working toward the PhD degree with the Department of Artificial Intelligence, Xiamen University, where he conducts research with the Media Analytics & Computing Laboratory. His primary research interests focus on deepfake detection, forgery detection, and camouflaged object detection. His work aims to develop advanced techniques in computer vision and deep learning to create robust systems that can effectively identify manipulated media, detect forged regions within images or videos, and localize camouflaged objects in complex environments. He serves as a reviewer for prestigious conferences and journals, such as ICCV, CVPR, ECCV, and *IEEE Transactions on Information Forensics and Security*.

Zhongxi Chen is currently working toward the dedicated graduate degree with the Institute of Artificial Intelligence of Xiamen University, Xiamen China, currently working within the Media Analytics & Computing Laboratory. His research is primarily concentrated on the domain of computer vision, with a particular emphasis on the detection of camouflaged objects, document tampering identification, and deepfake detection.

Xianming Lin received the PhD degree in intelligent multimedia information processing from the School of Informatics, Xiamen University, Xiamen, China, in 2014. He is currently an assistant professor with Xiamen University, Xiamen. His current research interests include visual retrieval, computer vision and machine learning.

Xiaoshuai Sun (Member, IEEE) received the BS degree in computer science from Harbin Engineering University in 2007, and the MS & PhD degree in computer science and technology with the Harbin Institute of Technology in 2009 and 2015 respectively. He is currently a professor with the School of Informatics, Xiamen University, China. His research interests include computer vision, pattern recognition, and multimedia analysis. He was a recipient of the best PhD thesis award in Harbin Institute of Technology (2015) and also a winner of the Microsoft Research Fellowship (2011). He owns 4 authorized patents and has authored more than 100 papers in referred journals and conferences, including *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *IEEE Transactions on Image Processing*, CVPR, ICCV, NeurIPS, IJCAI, AAAI and ACM MM.

Hong Liu received the PhD degree in computer science from Xiamen University in 2020. He is currently a specially-appointed assistant professor with Osaka University. His research interests include trustworthy AI and representation learning. He was awarded the Japan Society for the Promotion of Science (JSPS) International Fellowship, the Outstanding Doctoral Dissertation Awards of both the China Society of Image and Graphics (CSIG) and Fujian Province, the Top-100 Chinese New Stars in Artificial Intelligence by Baidu Scholar, and the Notable Reviewer at ICLR 2023. He serves as the associate editor of *Visual Intelligence*, the guest editor of IJCV, and the area chair of ACM MM.

Rongrong Ji (Senior Member, IEEE) is a Nanqiang distinguished professor with Xiamen University, the deputy director with the Office of Science and Technology, Xiamen University, and the director of Media Analytics and Computing Lab. He was awarded as the National Science Foundation for Excellent Young Scholars (2014), the National Ten Thousand Plan for Young Top Talents (2017), and the National Science Fundation for Distinguished Young Scholars (2020). His research falls in the field of computer vision, multimedia analysis, and machine learning. He has published 50+ papers in ACM/IEEE Transactions, including *IEEE Transactions on Pattern Analysis and Machine Intelligence* and IJCV, and 100+ full papers on top-tier conferences, such as CVPR and NeurIPS.